The Privacy-Fairness-Accuracy Frontier: A Computational Law & Economics Toolkit for Making Algorithmic Tradeoffs

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ABSTRACT

Both law and computer science are concerned with developing frameworks for protecting privacy and ensuring fairness. Both fields often consider these two values separately and develop legal doctrines and machine learning metrics in isolation from one another. Yet, privacy and fairness values can conflict, especially when considered alongside the accuracy of an algorithm. The computer science literature often treats this problem as an "impossibility theorem" - we can have privacy or fairness but not both. Legal doctrine is similarly constrained by a focus on the inputs to a decision - did the decisionmaker *intend* to use information about protected attributes. Despite these challenges, there is a way forward. The law has integrated economic frameworks to consider tradeoffs in other domains, and a similar approach can clarify policymakers' thinking around balancing accuracy, privacy, and fairnesss. This piece illustrates this idea by using a law & economics lens to formalize the notion of a Privacy-Fairness-Accuracy frontier, and demonstrating this framework on a consumer lending dataset. An open-source Python software library and GUI will be made available.

CCS CONCEPTS

• **Security and privacy** → *Human and societal aspects of security and privacy.*

KEYWORDS

privacy, fairness, consumer protection

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1 INTRODUCTION

Legal and computer science scholars have developed rich frameworks for addressing both privacy and fairness concerns in a wide variety of applications, yet there is little guidance about how to consider both values at once. The computer science literature often argues that fairness and privacy are in tension with one another, and presents the problem an "impossibility theorem" [5]. Similarly, the legal literature often sees privacy protections as an impediment to enforcing anti-discrimination law. Dwork and Mulligan identified this problem nearly a decade ago, yet the central policy issue persists [25]. Selbst and Barocas have similarly written extensively about the intersection of legal doctrines measuring discrimination and algorithmic decision-making [9]. In practice, treating the privacy/fairness tradeoff as intractable will leave policymakers ill-equipped to consider either one when making decisions. When thinking about algorithmic decisions, this problem becomes even more complicated as privacy and fairness must also be traded off against accuracy[1] [30]. This study introduces a law & economics approach to this problem, with a particular focus on applications to consumer protection law.

With regards to fairness, one major debate is how to reconsider legal standards for anti-discrimination lawsuits in light of algorithmic opacity. The law distinguishes between *disparate treatment* and *disparate impact* [13]. Disparate treatment requires that there was an intention to use information about a person's protected attribute when making a decision about them. Disparate impact requires showing that a practice has disparate outcomes for different subgroups, regardless of intention. The computer science literature provides several ways to measure disparate impact [29]. However, as models become more complex, scrutinizing inputs to models becomes difficult. This study follows Talia Gillis's suggestion to reframe consumer lending law toward "output scrutiny," and shows how to consider other values such as privacy alongside fairness [16].

Law and computer science have similarly co-evolved definitions of privacy. One example of this intersection is the growth of "differential privacy" as a popular concept [12, 14]. Differential privacy essentially limits how identifiable any individual could be given a query against a database. It is now being used in several government applications, including most recently by the U.S. Census to fuzzify data and promote access to researchers without compromising the privacy of vulnerable individuals [23].

Despite these challenges, there is a significant opportunity to construct a path forward. There is already an active area of research dealing with these tradeoffs in both law and computer science [5, 15, 20, 25, 26, 34]. Economic thinking has long pervaded other areas of law and given policymakers tools to consider tradeoffs in difficult decisions [11]. This piece integrates this tradition into the existing debates in law and computer science, and bridges the privacy-fairness gap by situating it within law & economics.

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[1]Note, this paper uses "accuracy" as shorthand for the performance of a machine learning algorithm. This choice is primarily aimed at ensuring simplicity and avoiding conflicting jargon between economics and machine learning. "Accuracy" should be read as standing in for the relevant machine learning metric (recall, precision, false positive parity, false negative parity, etc.)

Specifically, it introduces the idea of a "Privacy-Fairness-Accuracy" frontier, and argues that it can be integrated into pre-existing policy frameworks. Although this study focuses on the consumer lending context specifically, the central idea of building Privacy-Fairness-Accuracy frontiers can be extended to other areas of law as well.

Although collaborations between law and computer science have proven fruitful in operationalizing tools and definitions of fairness and privacy, there is relatively little work that integrates both together. It is important to address this gap because there are several high-stakes algorithmic decisions that will need to trade off between these concepts. More broadly, developing a regulatory framework for addressing privacy and fairness will be necessary to avoid the pitfalls of "privacy pretexts" [22] and the abuse of trade secrets law [32] to thwart attempts at algorithmic scrutiny. Firms that deploy unfair algorithms can shield themselves by claiming they are preserving privacy interests, or violate privacy protections under the guise of ensuring fairness. Developing standards for how analysts should consider objectives other than accuracy can help alleviate some of these concerns identified in the legal literature.

The main contribution of this piece is to closely integrate the legal and computer science literatures on this topic, and introduce a law & economics approach to addressing Privacy-Fairness-Accuracy tradeoffs. Law & economics provides a framework for considering the structure of law through the lens of efficiency and transaction costs [11]. By sketching a hypothetical Privacy-Fairness-Accuracy frontier, decision-makers can identify when strict efficiency improvements (meaning privacy, fairness, and accuracy can all be improved without sacrificing another value) are possible. Where strict improvements are not possible, lenders and policymakers can scrutinize the the relative winners and losers of a particular allocation between the three values and compensate the losers accordingly. The intuition of the frontier is simple - for any particular choice of algorithm, fairness criterion, and privacy standard, an analyst can visualize the tradeoffs. An important limitation is that the frontier is an empirical one rather than theoretical, meaning it exposes the tradeoffs for a particular set of choices on these three dimensions rather than the theoretical tradeoffs of all possible tradeoffs.

2 ALGORITHMIC PRIVACY, FAIRNESS, AND ACCURACY IN CONSUMER LENDING

This section lays out the theoretical frameworks that inform the overall policy problem. At a high level, consumer lending law balances lenders' incentives to make accurate credit risk determinations against consumers' privacy and civil rights. This section starts with brief overviews of law & economics, fairness in machine learning, and differential privacy for readers who may not be familiar with these areas. It then overviews the legal basis of consumer lending law. It concludes by discussing the tradeoffs between privacy, fairness, and Accuracy.

2.1 A Law & Economics Primer

This subsection outlines some key concepts from law & economics for unfamiliar readers. Law & economics has a long history within the legal academy and it would be difficult to summarize it adequately here, so readers are encouraged to explore further [11]. Broadly, law & economics is the application of microeconomic principles to the analysis of different areas of law such as contracts, torts, property, and constitutional law.

One central assumption in law & economics is the idea of **utility maximization.** In economics terms, utility refers to the value that an actor places on something. Individuals might find utility in happiness, voters might find utility in getting their policy preferences enacted, businesses might find it in profits, etc. Law & econ approaches assume that actors maximize their utility functions, which is a component of being considered a **rational actor**. A rational actor is one that, given a set of choices, can order those choices according to their own preferences and the ordering is transitive.[2] Utility maximization therefore implies that all rational actors are working in their own self-interest, meaning they wish to maximize the bundle of goods that gives them the most utility.

This concept lead to the core microeconomic tool that will motivate the analysis in this study: **efficiency**. One definition of efficiency is **productive efficiency** which has two conditions. First, it is not possible to produce the same amount of something using a lower-cost combination of inputs. Second, it is not possible to produce more of something with the same combination of inputs. More simply, productive efficiency means that production of a good is being maximized given current resources.

More central to law & economics theory, and to this study in particular, is **allocative efficiency**. Allocative efficiency is a concept that describes whether there are potential redistributions that maximize *aggregate* social utility across all actors. One popular concept to illustrate this idea is "Pareto efficiency," which is the idea that no changes can be made that can make someone better off without making someone else worse off.

Law & econ scholars will sometimes use a variant of Pareto efficiency known as Kaldor-Hicks efficiency that states efficiency is achieved when there are no moves where the winner of a reallocation could compensate the loser and still be at least as well off as before. This requirement is less stringent than Pareto efficiency as it allows for some parties to be made worse off, so long as they could hypothetically be compensated for their losses by the gains made by the winner.

This central insight from law & econ can help motivate how to regulate algorithmic credit decisions. A traditional law & econ approach would evaluate a legal rule's efficiency using a willingness-to-pay framework, meaning it quantifies each party's utility through money. This study extends framework to machine learning metrics. Efficiency can be seen as a policy value that is central to law & econ thinking. This insight is key to this study - efficiency can be seen as something like a "meta-value" that helps frame how to trade off between the other values under consideration. That being said, one future area of inquiry might be getting more traction on how to more comfortably fit fairness and privacy into the willingness-to-pay framework, or alternatively how to adapt regulation to incorporate qualitative notions into exercises such as cost-benefit analysis [27].

[2]Concretely, this means that given a bundle of goods A, B, and C a rational actor should be able to say they prefer A > B > C, and that they do not prefer C over A.

2.2 Fairness in Law and Machine Learning

Fairness in machine learning is a rapidly growing subfield with important stakes for a range of algorithmic decisionmaking contexts. Solon Barocas, Moritz Hardt, and Arvind Narayanan frame the problem around making decisions in "the right way and for the right reasons" [8]. Much of the scholarly attention to fairness in machine learning has focused on how algorithmic decisionmaking systems trained on biased data can reproduce historical disparities. For instance, in the consumer lending context, a loan-origination algorithm trained on historic mortgage lending data might flag Black applicants as uncreditworthy because of historic redlining practices. Scholars in this area have also developed a series of fairness metrics to measure different kinds of disparities in both the underlying training data and outcomes of algorithmic decisionmaking.

This study focuses on fairness in the outputs of algorithmic lending decisions. Following the approach used by the authors of the Aequitas Bias and Fairness Audit Toolkit, the core idea is to evaluate the fairness of the impact of a decisionmaking [29]. The major innovation of the Aequitas approach is that it does not presuppose that there is one fairness metric that is appropriate for every context, but rather situates the relevant metric within the decision being made. Are interventions punitive or assistive? Would false positives help or hurt the subject of a decision? Are errors distributed in a disproportionate way among relevant subgroups? Asking these sorts of questions can help clarify how to quantify the relevant fairness criterion. The authors introduce the "fairness tree" to help guide policymakers in understanding what metric would be best suited to a particular algorithmic decisionmaking context.[3]

2.3 Differential Privacy and Consumer Lending

Privacy as a policy value is expressed in many different ways, some of which are more amenable to a quantitative framework than others. Within the law, there are both constitutional and statutory privacy protections in different sectors of social life. Within consumer finance, the main focus of this study, the major pieces of legislation are the Gramm-Leach-Bliley Act (GLBA) [1] and the Fair Credit Reporting Act (FCRA) [3]. GLBA's approach to privacy protections rely on a notice-and-consent framework where financial institutions have an affirmative duty to disclose changes to privacy policies to consumers. Financial institutions also have an obligation not to disclose non-public data about applicants to third parties. The FCRA defines legally permissible purposes that data may be used for including furnishing consumer credit reports for certain transactions and complying with federal investigations.

Differential privacy is a framework that is gaining traction in legal applications. Differential privacy is a mathematical guarantee against *privacy attacks*, meaning attempts to learn information particular to individuals. First proposed by Cynthia Dwork, [14] tools that implement the concept have been growing in popularity. At a basic level, a computation can be considered differentially private if an outside observer cannot tell that a particular individual's data was used in that computation. Another way of framing this concept is that an outside observer will draw the same inference about an individual's information whether or not that individual's information was included as an input to the analysis [33]. Importantly, differential privacy is a property of the *analysis* and not its output [33]. The basic guarantee of differentially private analyses is that it is possible to reason about the overall privacy loss even after many differentially private computations are carried out.

The main benefit of differential privacy over other frameworks is that it is easily quantifiable. The primary measure of interest is the risk of an observer successfully inferring information about an individual in a dataset. This parameter is conceptualized as the privacy loss, ϵ. We can think of ϵ as a description of the "privacy budget" in an analysis. Each analysis or computation of a dataset poses a risk of exposing an individual's information. ϵ can be thought of as the overall privacy budget that bounds how much risk to an individual's privacy is tolerable across several analyses. $\epsilon = 0$ implies that an individual's information is not used in the computation at all. Lower values correspond to lower risk to individuals, and higher values correspond to higher risk. There is no universally correct choice of epsilon, and the choice will depend on the context and normative consideration. As a general rule, epsilon values ranging between .01 and 1 are suggested as good starting points, with .1 being seen as a good baseline protection [33].

Differential privacy is also a useful starting point for the toolkit because it is being proposed as a solution in various credit risk modeling algorithms [35]. Maniar et. al. propose a differentially private model for modeling both a consumer's probability of default and exposure (the amount of the loan balance that is outstanding) at time of default [24].

The main results in this paper utilize IBM's diffprivlib for modeling the differential privacy of a particular algorithm [17]. There are a few reasons for this particular choice. First, the IBM library is built with government use cases, such as the U.S. Census dataset releases, in mind. Second, the library is amenable to scoring the privacy loss of a variety of machine learning algorithms that are likely to be used in credit risk modeling, such as tree-based methods, logistic regression, etc. That being said, this choice is meant to be illustrative of representative use cases - a particular agency or application may require a different conceptualization of differential privacy that could be easily plugged into the toolkit.

2.4 Consumer Lending Law

Consumer law is a natural place to start examining the Privacy-Fairness-Accuracy framework because at the highest levels it has long focused on these values. Legislation like the Truth in Lending Act [2], Fair Credit Reporting Act [3], Equal Credit Opportunity Act [4], and Gramm-Leach-Bliley Act [1] balance between lending institutions desire to make accurate predictions about creditworthiness and consumer rights, including privacy. These laws, in addition to ones like the Fair Housing Act and Home Mortgage Disclosure Act are also concerned with addressing discrimination based on protected attributes. Few sectors are as heavily regulated as consumer finance, and there are several legal doctrines that map nicely to machine learning metrics. Critically, there are several disclosure requirements that make consumer lending data available to the government and researchers, making this toolkit valuable for a variety of consumer lending applications.

[3]For a helpful overview, see: http://www.datasciencepublicpolicy.org/our-work/tools-guides/aequitas/

One of the general themes from scholarly analysis of these laws is the tension between allowable and non-allowable information in decision-making. Banks and financial institutions are not permitted to use protected attributes when making consumer lending decisions. Yet, there remains an important question of how decision-making can be based on protected attributes in subtle, implicit ways. To illustrate, the Equal Credit Opportunity Act forbids discrimination based on a protected attribute, such as race, and allows credit decisions to only be made based on an applicant's creditworthiness. Yet, calculations about creditworthiness can include correlates with race. For instance, mortgage lending decisions can take into account projected property tax liability and homeowners insurance rates, both of which can highly correlate with race because of historical redlining.

As more consumer credit decisions are made algorithmically, a few key concerns emerge. One major problem with this approach is if the datasets used to train algorithms, or human decision-makers, to distinguish creditworthy from uncreditworthy applicants are biased, then decisions will lead to disparate outcomes. For instance, if Black applicants were historically deemed uncreditworthy because of taste-based discrimination, an algorithm that is trained on correlates of race like the applicant's zip code or education status will replicate the same bias.

2.5 The Privacy-Fairness-Accuracy Tradeoff

Both the law and computer science literatures have identified tradeoffs between the privacy, fairness, and accuracy of algorithmic decision-making. One of the main issues with considering all three metrics together is that there are usually tradeoffs between any pair of them [5]. Several studies within the computer science literature have investigated and quantified these tradeoffs. For example, there is evidence that prioritizing group fairness will lead to information leakage and therefore diminish privacy.[10] The converse of this idea is that prioritizing differential privacy can increase group disparities, and exacerbate group disparities to a larger extent than non-DP algorithms [7]. There are a variety of promising efforts to solve this problem by using differentially private and fair deep learning algorithms [26, 34].

Within the legal literature, there are similar debates about the tension between privacy and fairness, generally framed through the lens of anti-discrimination law. One argument is that privacy can harm anti-discrimination law by shielding relevant information from decision-makers, and thus incentivizing them to rely on group proxies for decision-making instead of an individual's relevant information [31]. A counterargument is that privacy can actually strengthen anti-discrimination by shielding information about protected attributes from decision-makers [28].

This study aims to make a modest addition to this literature by situating these debates within law & economics. Law & economics has already been used to inform consumer lending regulation [19]. The advantage of this approach is that it brings the traditional law & economics scholarship on consumer lending in closer conversation with studies of the algorithmic lending. More broadly, it should also highlight ways that law & econ can deal with algorithmic decisionmaking more generally.

3 POLICY FRAMEWORK FOR THE PRIVACY-FAIRNESS-ACCURACY FRONTIER

The main contribution of this study to develop a computational toolkit for assessing the Privacy-Fairness-Accuracy tradeoff. Reorienting the law from input scrutiny to output scrutiny will help harmonize legal and computational approaches to privacy and fairness.

3.1 Designing the Toolkit

This toolkit combines existing open source applications to furnish analysts with an easy way to evaluate the Privacy-Fairness-Accuracy frontier for a particular lending application, and scrutinize whether there are any efficiency improvements possible. At a high level, the toolkit does the following:

- Preprocesses data for preparation for machine learning
- Provides options for fitting logistic regression, Naive Bayes, and random forest models
- Implements hyperparameter grid search via grid search to find the theoretical "best" model, as specified by the user's chosen Accuracy metric (accuracy, precision, recall, etc.). The accuracy metric for this model will be considered the baseline for the best model without taking privacy or fairness into account.
- Adds a hyperparameter to tune differential privacy settings, and return a grid for the accuracy-privacy tradeoff
- Evaluates the fairness metrics at each level of differential privacy

The toolkit draws on three primary Python libraries. The scikit-learn library is used to hyperparameter tune and train the "best" models that do not take differential privacy or fairness into account. The diffprivlib library is used to tune differential privacy values [18]. The aequitas library is used to generate fairness metrics and return the analyst's choice of metric [29].

The main advantage of using this toolkit as opposed to working with each of these libraries separately is that it provides methods for modeling the Privacy-Fairness-Accuracy frontier of a particular algorithmic context. An important disclaimer is that this frontier exposes the tradeoffs between the choices of a) a particular decision (e.g. whether to extend a mortgage, whether to refinance, etc.) b) a particular machine learning algorithm (e.g. a random forest, a neural network etc.) and c) a privacy metric (e.g. differential privacy, knn privacy etc.) It also makes customizability easier for non-experts - a policymaker can choose different model configurations, differential privacy levels, and fairness metrics to model frontiers for different policy settings.

3.2 Evaluating Tradeoffs with Allocative Efficiency

After running the toolkit on credit (or other) data, the analyst can then evaluate tradeoffs using a familiar law & economics framework. If there are any reallocations of accuracy, privacy, and fairness that would represent a Kaldor-Hicks improvement, then this would be an obvious case for the lending institutions to recalibrate their algorithmic decision-making. The clearest example of this idea is

when a lending institution uses a non-differentially private algorithm and does not make adjustments for fairness, but also chooses an algorithm that is not optimal for accuracy (i.e. they might use a logistic regression instead of a neural network).

Once an algorithmic system is at the frontier, policymakers can engage with how to manage tradeoffs between the three values. Perhaps the law would prefer to prioritize anti-discrimination goals over privacy ones, say in cases like housing discrimination. In other cases, the law may prefer to choose a more privacy-protecting algorithm, say in cases dealing with national origin or citizenship. These choices are inherently complicated and involve political and ethical considerations so no one rule will be universally satisfactory. That being said, making the tradeoffs clear via allocative efficiency makes for a good starting place to start considering these questions.

4 DEMONSTRATION ON HMDA DATA

4.1 Data

This study uses data from the Home Mortgage Disclosure Act (HMDA) from 2007-2017. This dataset is available from the Consumer Financial Protection Bureau (CFPB), and contains the required disclosures for all home mortgage loans applied for and originated in the United States. The HMDA requires that financial institutions report whether a loan was originated or another action was taken, as well as other key information like the applicants race, gender, and income. The data also includes population-level data such as minority population proportion at the census tract level. There are approximately 10-12 million observations per year, amounting to over 100 million records overall. The main outcome of interest for this study is whether a loan is originated or not. Loans may not be originated for a number of reasons including being denied for insufficient credit rating, or the application being withdrawn by the applicant. The dataset is further preprocessed to exclude any features that would be perfectly predictive of whether a loan is originated (for example, if there are values in the "reason for denial" field).

4.2 Example Privacy-Fairness-Accuracy Frontier

Following on Gillis's suggestion about how to implement output scrutiny, this section presents a toy example. Gillis suggests that an algorithm should treat like cases alike - meaning that a credit score should mean the same thing for two similarly situated people. The Boston Federal Reserve branch conducted a study in 1990 arguing that there was evidence of racial discrimination in mortgage lending using the same definition [6]. Using race as an example, this idea is probably best conceptualized as False Negative/Group Size parity according to the Aequitas fairness tree. FN/GS is essentially a measure of how many people in a subgroup were erroneously denied a loan, scaled by the number of individuals in that group. Formally[29]:

$$\frac{FN}{GS} = P(\hat{Y} = 0, Y = 1|G)$$

Where:

- FN: False Negatives
- GS: Group Size
- $\hat{Y}$: Predicted label (loan originated 1, loan not originated 0)
- Y: True label
- G: Group membership

We can compare FN/GS fairness against *epsilon*-differential privacy [14] and an accuracy metric. Figure **??** shows one result of Gillis's and the Boston Fed's fairness metric against these two measures.

Of course, there are several different metrics that could be appropriate. A metric that is closer to the legal definition of disparate impact might be implementing something like the 80-20 rule. The 80-20 rule is a heuristic in employment anti-discrimination law that basically says that members of protected groups should not be hired at rate less than 80% of white men. We can apply this idea to the credit dataset by asking whether members of a protected group were granted loans at an acceptable rate relative to a reference group. We can formalize this fairness metric as:

$$Disparity = \frac{\frac{TP_{G1}}{GS_1}}{\frac{TP_{G2}}{GS_2}}$$

Where:

- TP: True Positives
- G: Group membership
- GS: Group size

Figure **??** shows this hypothetical frontier for women borrowers relative to men. The red region corresponds to disallowed combinations assuming strict adherance to the 80/20 rule. Therefore, allowable algorithms fall in the blue region, and policymakers could mandate an algorithm be chosen from the blue region, maximizing either accuracy or privacy to achieve Kaldor-Hicks efficiency. If policymakers wish to impose privacy constraints as well, say that ϵ differential privacy by at least 6, that would further narrow the region.

This approach can also illuminate where algorithms fail altogether. Conducting the same exercise on race, and consistent with the Boston Fed's findings, there are *no* algorithms in this test case that meet the fairness criterion for Black borrowers relative to white ones. If we relax the 80/20 rule to instead be 60/40, there are a few acceptable algorithms, as seen in Figure **??**. However, under the original 80/20 rule, the entire region would be red, suggesting a scenario where there are biases so deep that algorithmic decision-making is unlikely to correct them.

5 DISCUSSION, LIMITATIONS, AND FUTURE WORK

This study makes a number of contributions to the intersection of law, computer science, and economics. First, it presents a computational toolkit that regulators and scholars can use to easily evaluate Privacy-Fairness-Accuracy tradeoffs. Second, it bolsters existing calls for the law to move away from input scrutiny and toward output scrutiny, and provides a proof-of-concept for how regulators such as the CFPB and HUD can engage in data-driven output scrutiny. Finally, it enriches the interdisciplinary enterprise of law and computer science by situating it within the familiar framework of law & economics. Computational economics is also an interdisciplinary field, yet there have been few efforts to unify

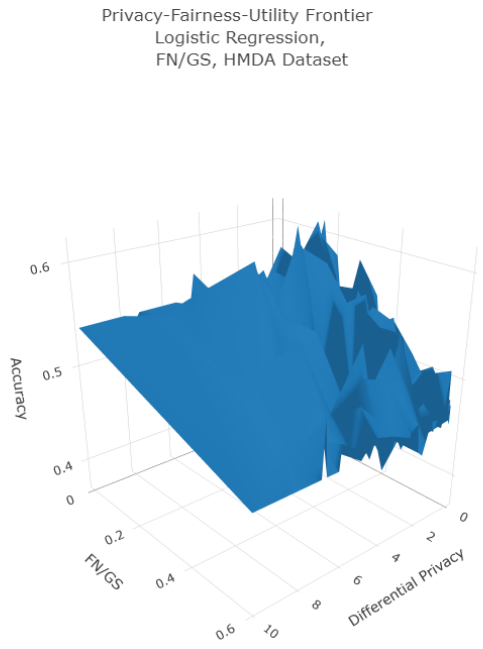

Figure 1: False Negative/Group Size Privacy-Fairness-Accuracy Frontier

Notes. Example frontier for different differential privacy and fairness settings for logistic regression trained on the HMDA dataset. In this case, accuracy and fairness tend to decrease as privacy increases. The key here is that there are Kaldor-Hicks improvements at several points (for example, at a FN/GS ratio of .2, differential privacy can be strengthened from $epsilon$ 4 to 2 to achieve an increase in accuracy of .1. Even at an ϵ-differential privacy value of 10, there are Pareto improvements from models with FN/GS of .3 and accuracy of .4 to FN/GS of 0 and accuracy of .53.

all three disciplines under one umbrella. This study represents a first step in how to integrate computational advances into law & economics, which has traditionally been more theory-driven. Computer scientists can also benefit from integrating law & economics frameworks into their own research. In particular, law & economics is useful for identifying the constraints that policymakers face, and future computer science research into these issues can benefit from attention to these real-world constraints.

Before implementing this toolkit in practice, it is important to recognize some important caveats. The first is that this study only considers accuracy, fairness, and privacy. Yet, there may be other values that are worth considering in algorithmic decisionmaking. There is a real danger of only including values that are easily operationalized in a quantifiable measure, and the scholarly community should be attentive to this issue. Furthermore, accuracy, fairness, and privacy can all be modeled in different ways than what is presented in this study. Different choices will be appropriate in different settings. This limitation also illustrates that the toolkit exposes the frontier of a particular set of choices, *not* the frontier for all possible combinations of these values in a global sense.

Law & economics is a powerful framework for illuminating tradeoffs, but does not solve the problem of redistribution by itself. Where strict improvements (for example, privacy can be increased without costs to accuracy or fairness) are possible, there is a clear case that regulators should compel lending institutions to make such improvements. However, the question of whether to privilege privacy, fairness, or accuracy at the frontier does not have a clear economic answer. Furthermore, while recent literature in law & economics has grappled with the question of how to engage in redistribution through the tax code, more work needs to be done in this space to understand what redistribution would mean in this context. If privacy protections disproportionately hurt Black or Latino borrowers, how should they be compensated for those harms? If the harms cannot be easily quantified in monetary terms, such compensation will be more difficult under current legal theories. This challenge will require reimagining the role of law and regulation, and should be an area of future work in law and computer science.

It is also important to recognize that this framework and toolkit work primarily because there are quantifiable metrics of privacy and fairness, but the law and computer science communities should still be attentive to the way values are embedded in metrics. Just because differential privacy and certain definitions of fairness can be measured does not mean they are the right definitions in every situation. Within law & economics, prioritizing efficiency is a value judgment as well. Efficiency is not necessarily an end goal

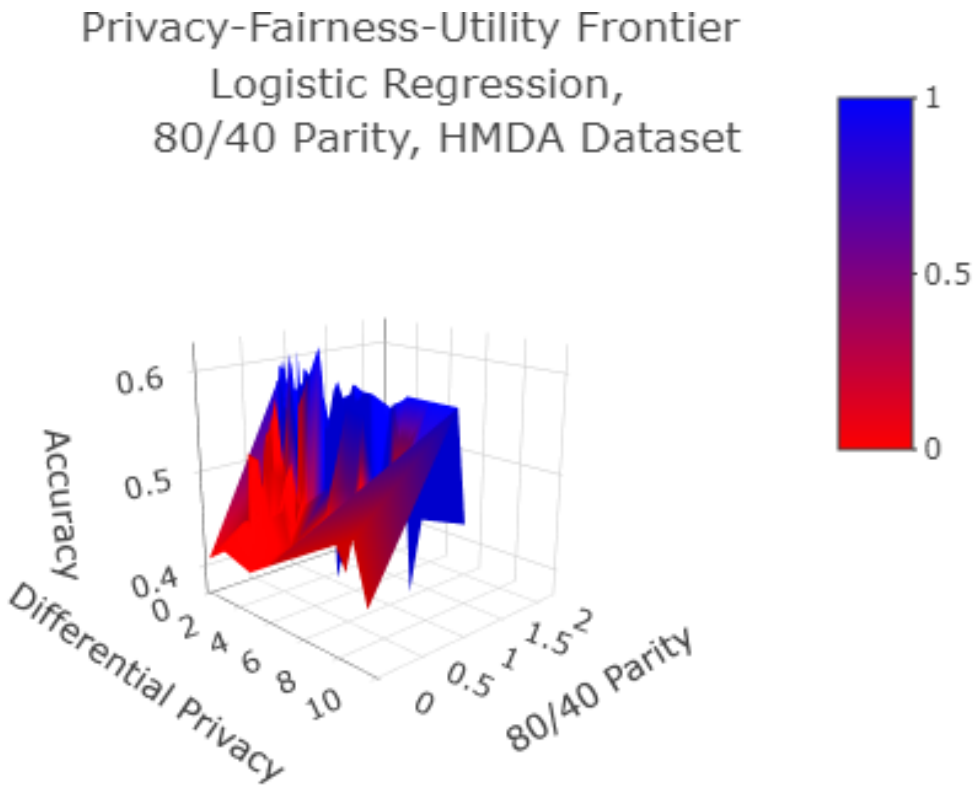

Figure 2: 80/20 Parity for Female and Male Borrowers Privacy-Fairness-Accuracy Frontier.

Notes. Example frontier for women borrowers using a 80/20 rule. Blue zones indicate decisions that comply with an 80/20 rule. Red zones indicate decisions that indicate invalid decisions under an 80/20 rule.

in and of itself. Kaldor-Hicks efficiency only requires that the winners *could* compensate the losers, not that they must. Chasing efficiency while ignoring lessons from the legal and computer science communities about building ethical frameworks could lead to exacerbating inequalities [21]. This effort to introduce a law & economics framework to privacy and fairness is intended to enrich an existing conversation, not supplant it. It represents one way to clarify tradeoffs and assist with regulatory monitoring of algorithmic decisions, but by no means replaces the larger conversation around how to develop ethical frameworks for preserving privacy and fairness.

Going forward, this toolkit will be improved for public use in several ways. First, it will incorporate differentially private and fair deep learning algorithms that have become more popular in the computer science literature. More demonstrations on other consumer lending data sources besides the HMDA such as the CFPB's auto-loan dataset will be added to further explore Privacy-Fairness-Accuracy tensions in different applications. Finally, in addition to the existing Python implementation, an interactive dashboard version will be made available. This will allow users to upload a dataset, choose a fairness metric that is matched with a particular legal doctrine, and generate an interactive 3D plot to explore the Privacy-Fairness-Accuracy frontier. This feature will allow users to easily investigate algorithmic models and identify strict efficiency improvements, or potential redistributive moves to change the balance between the three objectives. All software will be open sourced.

6 CONCLUSION

The Privacy-Fairness-Accuracy tradeoff presents both legal and computational problems, but leveraging a computational law & economics toolkit can help unify different disciplines. By situating machine learning metrics in legal doctrines, and applying a familiar analytical framework, policymakers and scholars can be better positioned to evaluate algorithmic decision-making. This study is primarily concerned with privacy and fairness in consumer law, but

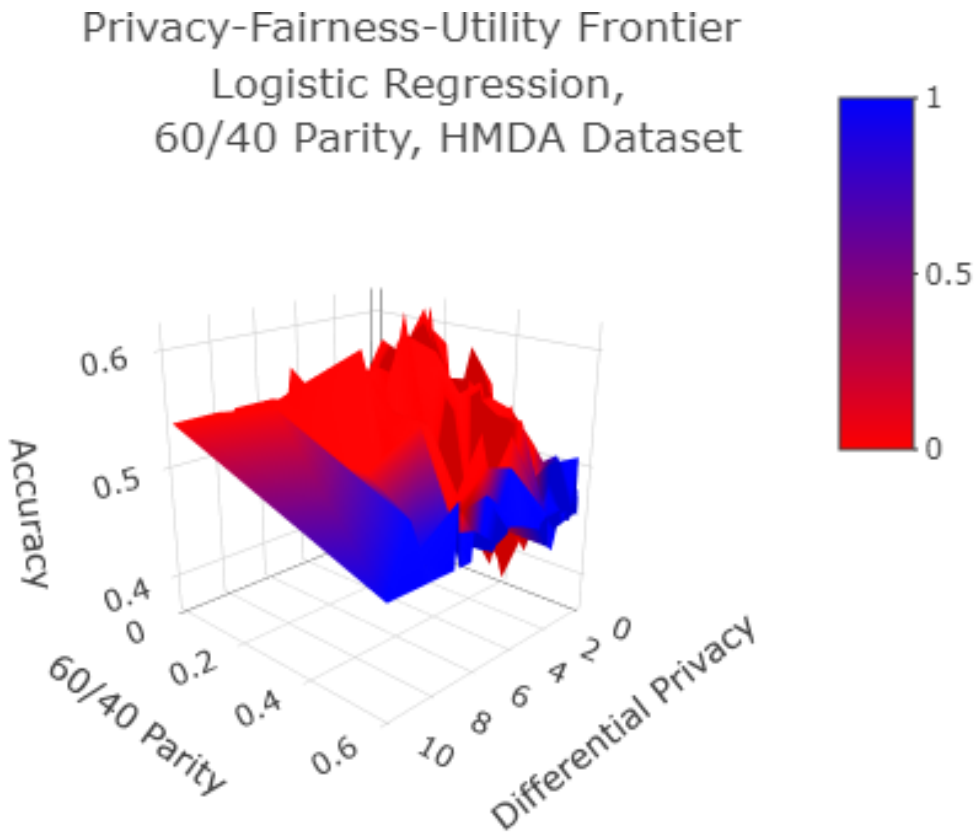

Figure 3: 60/40 Parity for Black and White Borrowers Privacy-Fairness-Accuracy Frontier

Notes. Example frontier for Black borrowers using a 60/40 rule. Blue zones indicate decisions that comply with an 60/40 rule. Red zones indicate decisions that indicate invalid decisions under an 60/40 rule.

can be extended to other domains that implicate the intersection of law and computer science.

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